

Orthonormal Facet Triad Foundations in Spherical Discrete Global Grid Systems: Radial Midpoint Unit Vector Formulation for Conservative Geospheric Fluxes

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Abstract

Discrete Global Grid Systems (DGGS) mapped to spherical topologies provide a uniform, partition-based discretization for planetary-scale thermodynamic modeling. In finite-volume formulations on hexagonal grids, such as Uber H3, accurately determining directional fluxes across inter-cell boundaries requires an orthonormal basis $(\hat{\mathbf{t}}, \hat{\mathbf{n}}_{\text{lat}}, \hat{\mathbf{n}}_{\text{rad}})$ defined at cell facets. This paper presents the mathematical specification, singularity mitigation, and implementation of `computeBoundarySegmentRadialNormal3D` within the Web of Life computational framework. The function extracts the normalized radial unit vector $\hat{\mathbf{n}}_{\text{rad}} \in \mathbb{S}^2$ passing through the Euclidean midpoint of boundary segments defined by coordinate pairs in $\mathbb{R}^3$. We prove that on any spherical shell, $\hat{\mathbf{n}}_{\text{rad}}$ is strictly orthogonal to the boundary segment chord tangent $\hat{\mathbf{t}}$, eliminating non-physical shear cross-talk in advective and radiative exchanges. We frame this development within finite-volume conservation laws, demonstrating how metric consistency prevents spurious entropy generation and mass-energy leakage in planetary transport simulations.

1 Introduction

Global thermodynamic and ecological modeling on spherical planetary manifolds demands discrete representations that minimize spatial distortion while respecting physical conservation principles [1]. Discrete Global Grid Systems (DGGS) employing hexagonal cell tiling, such as H3, offer quasi-uniform cell geometry and uniform neighborhood topology. However, when evaluating physical transport phenomena—such as barotropic fluid advection, latent and sensible heat transfer, and solar radiative forcing—continuous vector fields must be projected onto cell boundaries.

At every boundary facet shared by adjacent discrete cells, finite-volume balance equations rely on surface integrals of the flux tensor $\mathbf{F}$:

$$\oint_{\partial\Omega} \mathbf{F} \cdot \hat{\mathbf{n}} dS = \sum_{k=1}^K \int_{S_k} \mathbf{F} \cdot \hat{\mathbf{n}}_k dS \quad (1)$$

For polygonal segments S_k , resolving the local components requires an orthonormal coordinate triad:

1. The along-edge unit tangent vector $\hat{\mathbf{t}}$.
2. The outward radial unit normal vector $\hat{\mathbf{n}}_{\text{rad}}$, directed from the planetary geocenter through the boundary facet midpoint.

3. The lateral unit normal vector $\hat{\mathbf{n}}_{\text{lat}} = \hat{\mathbf{t}} \times \hat{\mathbf{n}}_{\text{rad}}$, lying tangent to the sphere and normal to the boundary edge.

Sprint 062 of the Web of Life project (<https://github.com/pascalranoroarijaona/WebOfLife>) implements the foundational geometric operator `computeBoundarySegmentRadialNormal3D` in `src/spatial/h3_adjacency.ts`. This manuscript establishes the formal mathematical derivations, singularity handling, thermodynamic invariance properties, and numerical verification of this formulation.

2 Mathematical Formulation

2.1 Segment Midpoint and Radial Unit Vector

Let $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{R}^3$ denote the Cartesian endpoints of a cell boundary segment on or near a sphere of radius R :

$$\mathbf{v}_1 = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} \quad (2)$$

The chord midpoint vector $\mathbf{m}$ connecting the planetary center $\mathbf{O} = [0, 0, 0]^T$ to the segment is:

$$\mathbf{m} = \frac{1}{2}(\mathbf{v}_1 + \mathbf{v}_2) \quad (3)$$

The unnormalized sum vector $\mathbf{s} = \mathbf{v}_1 + \mathbf{v}_2 = 2\mathbf{m}$ shares the identical directional ray. The radial normal unit vector $\hat{\mathbf{n}}_{\text{rad}}$ is defined as:

$$\hat{\mathbf{n}}_{\text{rad}} = \frac{\mathbf{s}}{\|\mathbf{s}\|_2} = \frac{\mathbf{v}_1 + \mathbf{v}_2}{\sqrt{(x_1 + x_2)^2 + (y_1 + y_2)^2 + (z_1 + z_2)^2}} \quad (4)$$

where $\|\cdot\|_2$ denotes the standard Euclidean norm in $\mathbb{R}^3$.

2.2 Orthogonality to Edge Tangent

Consider the directed chord tangent vector $\mathbf{t}_{\text{vec}} = \mathbf{v}_2 - \mathbf{v}_1$. On a spherical shell where $\|\mathbf{v}_1\|_2 = \|\mathbf{v}_2\|_2 = R$:

$$\mathbf{t}_{\text{vec}} \cdot \mathbf{s} = (\mathbf{v}_2 - \mathbf{v}_1) \cdot (\mathbf{v}_2 + \mathbf{v}_1) = \|\mathbf{v}_2\|_2^2 - \|\mathbf{v}_1\|_2^2 = R^2 - R^2 = 0 \quad (5)$$

Normalizing both vectors yields:

$$\hat{\mathbf{t}} = \frac{\mathbf{v}_2 - \mathbf{v}_1}{\|\mathbf{v}_2 - \mathbf{v}_1\|_2}, \quad \hat{\mathbf{n}}_{\text{rad}} = \frac{\mathbf{v}_1 + \mathbf{v}_2}{\|\mathbf{v}_1 + \mathbf{v}_2\|_2} \implies \hat{\mathbf{t}} \cdot \hat{\mathbf{n}}_{\text{rad}} = 0 \quad (6)$$

Thus, the radial normal through the Euclidean midpoint is unconditionally orthogonal to the boundary segment tangent for all spherical chords. The transverse lateral normal is subsequently defined via the cross product:

$$\hat{\mathbf{n}}_{\text{lat}} = \hat{\mathbf{t}} \times \hat{\mathbf{n}}_{\text{rad}} \quad (7)$$

completing a right-handed orthonormal triad $(\hat{\mathbf{t}}, \hat{\mathbf{n}}_{\text{lat}}, \hat{\mathbf{n}}_{\text{rad}})$ on $\mathbb{S}^2$.

2.3 Singularity Mitigation and Epsilon Guard

In computational geometry, floating-point evaluation of $\|\mathbf{s}\|_2$ can encounter severe loss of significance or division by zero if endpoints are antipodal ($\mathbf{v}_1 = -\mathbf{v}_2$) or degenerate ($\mathbf{v}_1 = \mathbf{v}_2 = \mathbf{0}$).

To prevent non-deterministic IEEE-754 NaN propagation, an invariant threshold $\epsilon = 10^{-12}$ is enforced:

$$\hat{\mathbf{n}}_{\text{rad}} = \begin{cases} \frac{\mathbf{s}}{\|\mathbf{s}\|_2}, & \text{if } \|\mathbf{s}\|_2 > \epsilon \\ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, & \text{if } \|\mathbf{s}\|_2 \leq \epsilon \end{cases} \quad (8)$$

In strict runtime configurations, an explicit domain exception is raised rather than falling back to the canonical zenith vector.

3 Thermodynamic Flux Decomposition

3.1 Conservation Principles

The operator `computeBoundarySegmentRadialNormal3D` performs a pure coordinate space transformation. It satisfies exact matter-energy conservation:

$$\Delta M = 0 \text{ kg}, \quad \Delta U = 0 \text{ J}, \quad \Delta S_{\text{universe}} \geq 0 \quad (9)$$

However, the orientation of $\hat{\mathbf{n}}_{\text{rad}}$ directly dictates the accuracy of projected thermodynamic transport rates across the cell perimeter.

3.2 Hydrodynamic and Thermal Facet Fluxes

For adjacent H3 cells A and B sharing an edge of length L_s and atmospheric depth Δz , the inter-cell face area is $A_f = L_s \Delta z$. Let $\mathbf{u} \in \mathbb{R}^3$ be the local fluid velocity. The lateral advection speed crossing the facet is:

$$u_{\text{lat}} = \mathbf{u} \cdot \hat{\mathbf{n}}_{\text{lat}} = \mathbf{u} \cdot (\hat{\mathbf{t}} \times \hat{\mathbf{n}}_{\text{rad}}) \quad (10)$$

The corresponding mass and enthalpy transfers over time interval Δt are:

$$\Delta M_{\text{H}_2\text{O}} = (u_{\text{lat}} A_f) \rho_{\text{air}} q_v \Delta t \quad (11)$$

$$\Delta H = (u_{\text{lat}} A_f) \rho_{\text{air}} c_p (T_A - T_B) \Delta t \quad (12)$$

where ρ_{air} is air density, q_v is specific humidity, c_p is isobaric specific heat capacity, and T is temperature.

3.3 Radiative Forcing and Zenith Angle Resolution

Boundary facet midpoints serve as quadrature points for incident solar irradiance. Let $\hat{\mathbf{s}} \in \mathbb{S}^2$ represent the solar illumination direction vector. The effective solar zenith angle θ_z evaluated at the segment midpoint is given by:

$$\cos \theta_z = \max(0, \hat{\mathbf{s}} \cdot \hat{\mathbf{n}}_{\text{rad}}) \quad (13)$$

The shortwave radiant energy absorbed along the facet boundary over time step Δt is:

$$\Delta E_{\text{solar}} = I_0 \tau_{\text{atm}} (\hat{\mathbf{s}} \cdot \hat{\mathbf{n}}_{\text{rad}}) A_{\text{facet}} \Delta t \quad (14)$$

where I_0 is the solar constant and τ_{atm} is atmospheric transmissivity. Accurate computation of $\hat{\mathbf{n}}_{\text{rad}}$ guarantees that solar insolation smoothly transitions to zero across the terminator without facet-misalignment artifacts.

4 Implementation & Architectural Integration

The routine is implemented in TypeScript within `src/spatial/h3_adjacency.ts`:

```
export function computeBoundarySegmentRadialNormal3D(
  segment: BoundarySegment3D,
  epsilon: number = 1e-12
): Vector3D {
  return computeBoundarySegmentRadialNormal3DFromPoints(
    segment.v1, segment.v2, epsilon
  );
}
```

The underlying function computes:

$$x_m = x_1 + x_2, \quad y_m = y_1 + y_2, \quad z_m = z_1 + z_2 \quad (15)$$

$$r = \sqrt{x_m^2 + y_m^2 + z_m^2} \quad (16)$$

If $r \leq \epsilon$, $[0, 0, 1]^T$ is returned; otherwise, $[x_m/r, y_m/r, z_m/r]^T$ is returned.

5 Numerical Experiments and Verification

To evaluate numerical stability, unit invariance, and orthogonality, a series of benchmark tests were executed.

Table 1: Benchmark results for `computeBoundarySegmentRadialNormal3D`.

Configuration	Endpoints ($\mathbf{v}_1, \mathbf{v}_2$)	Computed $\hat{\mathbf{n}}_{\text{rad}}$	Orthogonality Error $ \hat{\mathbf{t}} \cdot \hat{\mathbf{n}}_{\text{rad}} $
Equatorial Arc	$[(1, 0, 0)^T, [0, 1, 0]^T]$	$[0.707107, 0.707107, 0.0]^T$	$< 1.11 \times 10^{-16}$
Mid-Latitude Cap	$[(0, 1, 1)^T, [1, 0, 1]^T]$	$[0.408248, 0.408248, 0.816497]^T$	$< 1.11 \times 10^{-16}$
Polar Apex Chord	$[-0.5, 0, 1]^T, [0.5, 0, 1]^T]$	$[0.0, 0.0, 1.0]^T$	$< 1.11 \times 10^{-16}$
Homogeneous Dilation	$[2, 0, 0]^T, [0, 2, 0]^T]$	$[0.707107, 0.707107, 0.0]^T$	$< 1.11 \times 10^{-16}$
Antipodal Degeneracy	$[1, 0, 0]^T, [-1, 0, 0]^T]$	$[0.0, 0.0, 1.0]^T$ (Fallback)	N/A

As documented in Table 1, machine-precision orthogonality ($< 1.11 \times 10^{-16}$) is maintained across non-degenerate configurations. Under uniform scaling ($\mathbf{v} \mapsto \alpha \mathbf{v}$, $\alpha > 0$), the calculated normal remains invariant within machine epsilon.

6 Conclusion

Sprint 062 solidifies the spatial differential geometry layer of the Web of Life engine by deploying `computeBoundarySegmentRadialNormal3D`. By resolving the local vertical axis at boundary midpoints, the framework enables robust construction of the facet orthonormal triad $(\hat{\mathbf{t}}, \hat{\mathbf{n}}_{\text{lat}}, \hat{\mathbf{n}}_{\text{rad}})$. This capability provides the metric precision necessary for conservative advective flux projections, solar zenith irradiance modeling, and thermodynamic exergy dissipation tracking across planetary discrete global grids.

References

- [1] K. Sahr, D. White, and A. J. Kimerling, “Geodesic discrete global grid systems,” *Cartography and Geographic Information Science*, vol. 30, no. 2, pp. 121–134, 2003.

- [2] P. Ranoroarijaona, “Web of Life: Thermodynamic Earth System Architecture and Planetary Monad Engine,” *Open Source Codebase and Technical Specifications*, 2025. <https://github.com/pascalranoroarijaona/WebOfLife>.